MATH 152
Mrs. Bonny Tighe

EXAM I

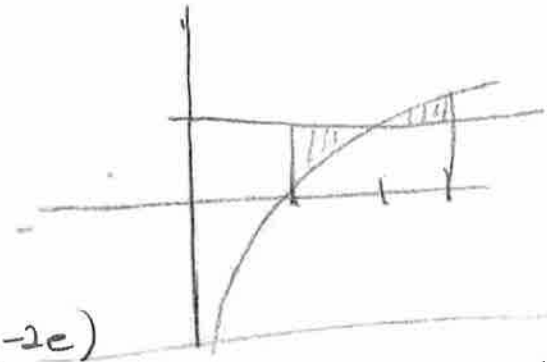
6.1-7.5
50 points
PART ONE

Name Solutions

TA_____ Wed 2/28/24

$\ln x = 1$
$e = x$

1. (10 points) Find the area of the region bounded the curves $y = \ln x$, *and* $y = 1$ on the interval $1 \le x \le 3$. Graph.



$\int_1^e (1-\ln x)\,dx + \int_e^3 (\ln x - 1)\,dx$

$x - (x\ln x - x)\Big|_1^e + (x\ln x - x - x)\Big|_e^3$

$2x - x\ln x\Big|_1^e + x\ln x - 2x\Big|_e^3$

$(2e - e(1)) - (2-0) + (3\ln 3 - 6) - (e - 2e)$

$2e - e - 2 + 3\ln 3 - 6 + e$

$e - 2$

$\boxed{2e + 3\ln 3 - 8}$

$\int \ln x\,dx$ $\quad u = \ln x \quad dv = dx$
$du = \frac{1}{x}dx \quad v = x$

$x\ln x - \int x\left(\frac{1}{x}\right)dx$

$x\ln x - x$

2. (2 points each) True or false Circle the correct True or False

True **False** a) $\tan^2 x + 1 = \sec^2 x + C$

True **False** b) $\dfrac{x^2+1}{x^2-4}$ can be put in the form $\dfrac{A}{x-2} + \dfrac{B}{x+2}$

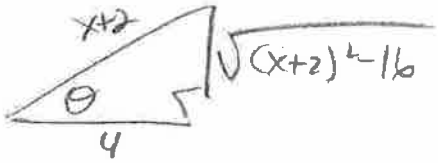
True False c) $\int \dfrac{y^2}{\sqrt{4y^2+16}}\,dy$, we need to use the trig substitution $y = 2\tan\theta$.

$2y = 4\tan\theta$

So $y = 2\tan\theta$

True False d) $\int_0^1 \frac{1}{x^2+1}dx$ should be evaluated using partial fraction decomposition.

True False e) If $\sec\theta = \frac{(x+2)}{4}$, then $\sin\theta = \frac{\sqrt{(x+2)^2-16}}{4}$



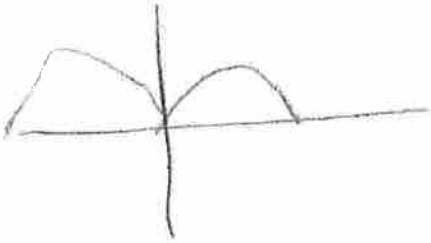
True False f) To evaluate $\int \cos^2(w)dw$ you need to use u-substitution.

True False g) An average value can only be a positive value .

True False h) $\int x^2 \sec(x^3)\,dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2}+\frac{C}{x+1}$

True False j) The Washer Method can be used to find the volume of the solid of revolution for the function $y = 4x - x^2$ about the $y-axis$.



3. (5 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^5+x^4}$ have.

$x^4(x+1)$

A) 4

B) 6

C) 3

D) 5

E) Some other answer

b) What method should not be used to integrate $\int \frac{x}{x^2-4}dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x)=2x-6x^2$ on the interval $[0,b]$ is -3, what is b?

$\frac{1}{b}\int_0^b (2x-6x^2)dx = \frac{1}{b}\left[x^2-2x^3\right]_0^b$

$\frac{1}{b}\left[b^2-2b^3\right] = b-2b^2 = -3$

$0 = 2b^2-b-3$

$(2b-3)(b+1)$

$b=3/2 \quad \cancel{b=-1}$

A) 3

B) $+1$

C) $3/2$

D) 0

E) Some other answer

d) After making a trigonometric substitution for $\int \frac{x^2\,dx}{\sqrt{16+x^2}}$

$x = 4\tan\theta$

$dx = 4\sec^2\theta\, d\theta$

$\int \frac{16\tan^2\theta\,(4\sec^2\theta)\,d\theta}{\sqrt{16+16\tan^2\theta}}$

$\int \frac{16\tan^2\theta\, 4\sec^2\theta\, d\theta}{4\sec\theta}$

$\int 16\tan^2\theta \cdot \sec\theta\, d\theta$

A) $\int 16\sin^2\theta\, d\theta$

B) $\int 16\tan^2\theta\sec^2\theta\, d\theta$

C) $\int 16\tan^2\theta\, d\theta$

D) $\int \frac{4\sin^2\theta}{\cos\theta}\, d\theta$

E) Some other answer

e) After using integration by parts one time on the integral $\int y3^y\, dy$, you would have:

$u = y$ $\quad dv = 3^y\, dy$

$du = dy$ $\quad v = \frac{3^y}{\ln 3}$

$y\left(\frac{3^y}{\ln 3}\right) - \int \frac{3^y}{\ln 3}\, dy$

A) $y3^y - \int 3^y\, dy$

B) $y3^y - \frac{1}{\ln 3}\int 3^y\, dy$

C) $\frac{y3^y}{\ln 3} - \frac{4^y}{\ln 3}$

D) $\frac{1}{\ln 3}y3^y - \frac{1}{\ln 3}\int 3^y\, dy$

E) Some other answer

MATH 152
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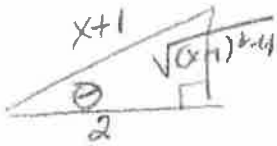
EXAM I

6.1-7.5
50 points
PART TWO

NAME Solutions

TA _____ Wed 2,



1. (10 points) Integrate $\int \frac{1}{\sqrt{x^2+2x-3}}\,dx$

$(x+1)^2-4$

$(x+1) = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

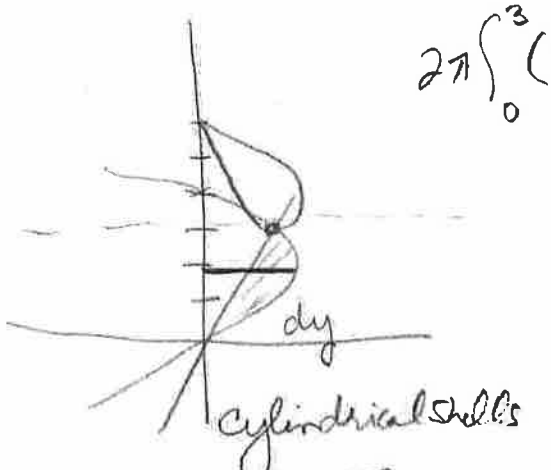
$$\int \frac{1}{\sqrt{(x+1)^2-4}}\,dx$$

$$\int \frac{2\sec\theta\tan\theta\,d\theta}{\sqrt{4\sec^2\theta-4}} = \int \frac{2\sec\theta\tan\theta\,d\theta}{\sqrt{4(\sec^2\theta-1)}} = \int \frac{2\sec\theta\tan\theta\,d\theta}{2\tan\theta}$$

$$\int \sec\theta\,d\theta = \ln|\sec\theta+\tan\theta|+C$$

$$\ln\left|\frac{x+1}{2}+\frac{\sqrt{(x+1)^2+4}}{2}\right|+C$$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $x = 4y - y^2$ *and* $y = x$ rotated about the line y = 4. Graph



cylindrical shells

$$2\pi\int_0^3 (4-y)[4y-y^2-y]\,dy = 2\pi\int_0^3 (4-y)[3y-y^2]\,dy$$

$$2\pi\int_0^3 (12y-4y^2-3y^2+y^3)\,dy$$

$$2\pi\int_0^3 (12y-7y^2+y^3)\,dy$$

$$2\pi\left[6y^2-\frac{7}{3}y^3+\frac{1}{4}y^4\right]_0^3$$

$$2\pi\left[6(9)-\frac{7}{3}(27)+\frac{1}{4}(81)-0\right]$$

$$2\pi\left[54-63+\frac{81}{4}\right]$$

$$2\pi\left[-9+20\tfrac{1}{4}\right] = 2\pi\left[11\tfrac{1}{4}\right] = 2\left(\frac{45}{4}\pi\right)$$

$$\frac{45\pi}{2}$$

if they had washers
2 - graph
2 - 4 - for radius
1 - 0 to 3

3. (10 points) Integrate: $\int e^{3t} \cos t \, dt$

$u = e^{3t}$	$dv = \cos t \, dt$
$du = 3e^{3t} dt$	$v = \sin t$

$$\int e^{3t} \cos t \, dt = e^{3t} \sin t - \int \sin t \, (3e^{3t}) \, dt$$
$$= e^{3t} \sin t - 3 \int \sin t \, e^{3t} \, dt$$

$u = e^{3t}$	$dv = \sin t \, dt$
$du = 3e^{3t} dt$	$v = -\cos t$

$$= e^{3t} \sin t - 3\left[e^{3t}(-\cos t) - \int -\cos t \, (3e^{3t}) \, dt\right]$$
$$= e^{3t} \sin t + 3e^{3t} \cos t - 9 \int \cos t \, e^{3t} \, dt$$

☺

$$\frac{1}{10}\left[e^{3t} \sin t + 3e^{3t} \cos t\right] + C$$

4. (10 points) Integrate: $\int_{\sqrt{3}}^{2} \frac{x^2}{\sqrt{4 - x^2}} \, dx$

$dx = 2\cos\theta \, d\theta$

$x = 2\sin\theta$ $\quad \sin\theta = x/2$

$\sin\theta = \sqrt{3}/2$ $\quad \sin\theta = 2/2 = 1$

$\theta = \pi/3$ $\quad \theta = \pi/2$

$$\int_{\pi/3}^{\pi/2} \frac{4\sin^2\theta \, 2\cos\theta \, d\theta}{\sqrt{4 - 4\sin^2\theta}}$$

$$\int_{\pi/3}^{\pi/2} \frac{4\sin^2\theta \, 2\cos\theta \, d\theta}{\sqrt{4(1 - \sin^2\theta)}} = \int_{\pi/3}^{\pi/2} \frac{4\sin^2\theta \, \cancel{2\cos\theta} \, d\theta}{\cancel{2\cos\theta}}$$

$$\int_{\pi/3}^{\pi/2} 4\sin^2\theta \, d\theta = \int_{\pi/3}^{\pi/2} (2 - 2\cos 2\theta) \, d\theta$$

$$2\theta - \sin 2\theta \Big]_{\pi/3}^{\pi/2} = \left(2(\pi/2) - \sin(\pi)\right) - \left(2(\pi/3) - \sin 2\pi/3\right)$$

$$\pi - 0 - 2\pi/3 + \sqrt{3}/2$$

$$\pi/3 + \sqrt{3}/2$$

$$x^4+x^2 \overline{\big)\, x^5 + 2} \quad \text{quotient } x$$
$$-(x^5+x^3+x)$$
$$-x^3+2$$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\,dx$

$$\int x + \frac{-x^3+2}{x^2(x^2+1)}\,dx$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} = \frac{-x^3+2}{x^2(x^2+1)}$$

$$Ax(x^2+1) + B(x^2+1) + x^2(Cx+D) = -x^3+2$$

$$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2$$

$$x^3(A+C) + x^2(B+D) + Ax + B = -x^3+2$$

$$A+C=-1 \quad D=-2 \quad A=0 \quad B=2$$

$$C=-1$$

$$\int\left(x + 2x^{-2} + \frac{-x-2}{x^2+1}\right)dx = \int\left(x + 2x^{-2} - \frac{x}{x^2+1} - \frac{2}{x^2+1}\right)dx$$

$$\frac{1}{2}x^2 - \frac{2}{x} - \frac{1}{2}\ln|x^2+1| - 2\tan^{-1}x + C$$

6. (5 points) Integrate $\int \sin\sqrt{x}\,dx$

$$m = \sqrt{x}$$

$$dm = \frac{1}{2\sqrt{x}}\,dx \qquad dx = 2m\,dm$$

$$\int 2m \sin m\,dm$$

$$u = 2m \qquad dv = \sin m\,dm$$

$$du = 2dm \qquad v = -\cos m$$

$$2m(-\cos m) - \int -\cos m\,(2dm)$$

$$-2m\cos m + 2\sin m + C$$

$$-2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x} + C$$

MATH 152
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EXAM IA

6.1-7.5
50 points
PART ONE

Name Solutions

TA_____ Wed 2/28/2024

1. (2 points each) True or false Circle the correct True or False

True False a) $\tan^2 x = \sec^2 x + C$

True False b) $\dfrac{x^2+1}{x^2-4}$ can be put in the form $\dfrac{A}{x-2}+\dfrac{B}{x+2}$

True False c) $\int \dfrac{y^2}{\sqrt{4y^2-16}}\,dy$, we need to use the trig substitution $y = 2\sec\theta$.

True False d) $\int_0^1 \dfrac{1}{x^2+4}\,dx$ should be evaluated using partial fraction decomposition.

True False e) If $\cos\theta = \dfrac{(x+2)}{4}$, then $\sin\theta = \dfrac{\sqrt{(x+2)^2-16}}{4}$

True False f) To evaluate $\int \sin^2(w)\,dw$ you need to use u-substitution.

True False g) An average value can only be a positive value .

True False h) $\int \sin x(\sin(\cos x))\,dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2}+\frac{C}{x+1}$

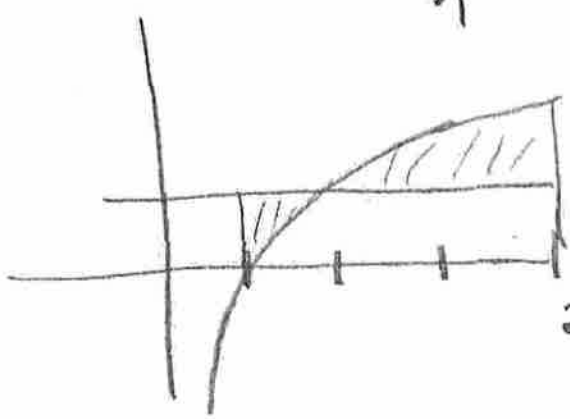
True False j) The Washer Method can be used to find the volume of the solid of revolution for the function $y = 4x - x^2$ about the $y-axis$.

2. (10 points) Find the area of the region bounded the curves $y = \ln x$, *and* $y = 1$ on the interval $1 \le x \le 4$. Graph.

$\ln x = 1$

$e = x$

$$\int_1^e (1-\ln x)\,dx + \int_e^4 (\ln x - 1)\,dx$$



$$x - (x\ln x - x)\Big|_1^e + (x\ln x - x - x)\Big|_e^4$$

$$2x - x\ln x\Big|_1^e + x\ln x - 2x\Big|_e^4$$

$$(2e - e\ln e) - (2-0) + (4\ln 4 - 8) - (e\ln e - 2e)$$

$$2e - e - 2 + 4\ln 4 + 8 - e + 2e$$

$$\boxed{2e + 6 + 4\ln 4}$$

$$\int \ln x\,dx = \boxed{\begin{matrix} u = \ln x & dv = dx \\ du = \frac{1}{x}\,dx & v = x \end{matrix}}$$

$$= x\ln x - \int x\left(\frac{1}{x}\right)dx$$

$$x\ln x - x + C$$

3. (5 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^6 - x^4}$ have.

$x^4(x^2-1)$

A) 4

B) 6

C) 3

D) 5

E) Some other answer

b) What method should not be used to integrate $\int \frac{x}{x^2 - 4} dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x) = 6x^2 + 2x$ on the interval $[0, b]$ is 3, what is b?

A) 3

B) $+1$

C) $+3/2$

D) 0

E) Some other answer

d) After making a trigonometric substitution for $\int \frac{x^2\,dx}{\sqrt{16-x^2}}$

A) $\int 16\sin^2\theta\,d\theta$

B) $\int 16\tan^2\theta\sec^2\theta\,d\theta$

C) $\int 16\tan^2\theta\,d\theta$

D) $\int \frac{4\sin^2\theta}{\cos\theta}\,d\theta$

E) Some other answer

e) After using integration by parts one time on the integral $\int y3^y\,dy$, you would have:

A) $y3^y - \int 3^y\,dy$

B) $y3^y - \frac{1}{\ln 3}\int 3^y\,dy$

C) $\frac{y3^y}{\ln 3} - \frac{4^y}{\ln 3}$

D) $\frac{1}{\ln 3}y3^y - \frac{1}{\ln 3}\int 3^y\,dy$

E) Some other answer

MATH 152
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EXAM IA

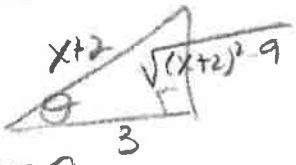
6.1-7.5
50 points
PART TWO

NAME Solutions

TA_____Wed 2/28/2024

1. (10 points) Integrate $\int \frac{1}{\sqrt{x^2+4x-5}}dx$

$(x+2)^2-9$



$(x+2) = 3\sec\theta$

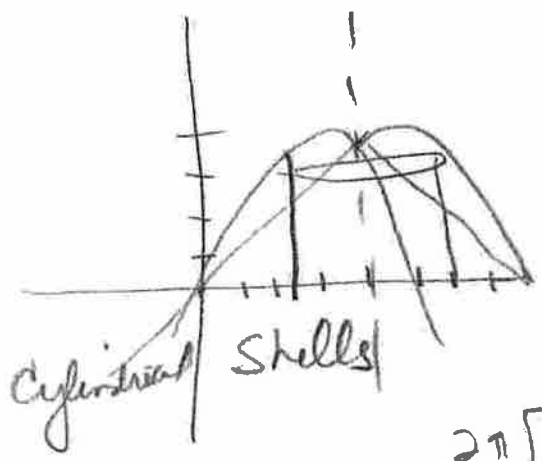
$dx = 3\sec\theta\tan\theta\, d\theta$

$$\int \frac{1}{\sqrt{(x+2)^2-9}}dx$$

$$\int \frac{3\sec\theta\tan\theta\, d\theta}{\sqrt{9\sec^2\theta-1}} = \int \frac{3\sec\theta\tan\theta\, d\theta}{3\tan\theta} = \int \sec\theta\, d\theta$$

$\ln|\sec\theta+\tan\theta|+C$ $\quad \ln\left|\frac{x+2}{3}+\frac{\sqrt{(x+2)^2-9}}{3}\right|+C$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $y = 5x - x^2$ *and* $y = x$ rotated about the line x = 4. Graph



Cylindrical Shells

$$2\pi\int_0^4 (4-x)[5x-x^2-x]\,dx$$

$$2\pi\int_0^4 (4-x)(4x-x^2)\,dx = 2\pi\int_0^4 (16x-4x^2-4x^2+x^3)$$

$$2\pi\int(16x-8x^2+x^3)\,dx =$$

$$2\pi\left[8x^2-\frac{8}{3}x^3+\frac{1}{4}x^4\right]_0^4 =$$

$$2\pi\left[2(4^3)-\frac{8}{3}(4)^3+4^3\right] =$$

$$2\pi(4^3)\left[2-\frac{8}{3}+1\right] = 128\pi\left(\frac{1}{3}\right) = \frac{128\pi}{3}$$

3. (10 points) Integrate: $\int e^t \sin 3t \, dt$

$u = \sin 3t \quad dv = e^t dt$

$du = 3\cos 3t \, dt \quad v = e^t$

$$e^t \sin 3t - \int e^t 3\cos 3t \, dt = e^t \sin 3t - 3\int e^t \cos 3t \, dt$$

$u = \cos 3t \quad dv = e^t dt$

$du = -3\sin 3t \, dt \quad v = e^t$

$$e^t \sin 3t - 3\left[e^t \cos 3t - \int e^t(-3\sin 3t)\,dt\right]$$

$$e^t \sin 3t - 3e^t \cos 3t - 9\int e^t \sin 3t \, dt$$

$$\frac{1}{10}\left[e^t \sin 3t - 3e^t \cos 3t\right] + C$$

4. (10 points) Integrate: $\int_{\sqrt{3}}^{2} \frac{x^2}{\sqrt{4-x^2}}\,dx$

$x = 2\sin\theta \quad \sin\theta = x/2$

$dx = 2\cos\theta \, d\theta \quad \sin\theta = 2/2 = 1$

$\sin\theta = \sqrt{3}/2 \quad \theta = \pi/2$

$\theta = \pi/3$

$$\int_{\pi/3}^{\pi/2} \frac{4\sin^2\theta}{\sqrt{4-4\sin^2\theta}} 2\cos\theta \, d\theta$$

$$\int_{\pi/3}^{\pi/2} \frac{4\sin^2\theta \; 2\cos\theta \, d\theta}{2\cos\theta} = \int_{\pi/3}^{\pi/2} 4\sin^2\theta \, d\theta =$$

$$\int_{\pi/3}^{\pi/2} (2 - 2\cos 2\theta)\,d\theta = 2\theta - \sin 2\theta \Big|_{\pi/3}^{\pi/2}$$

$$\left(2(\pi/2) - \sin\pi\right) - \left(2(\pi/3) - \sin 2\pi/3\right)$$

$$\pi - 0 - 2\pi/3 + \sqrt{3}/2$$

$$\pi/3 + \sqrt{3}/2$$

5. (10 points) Integrate $\int \frac{x^6+2}{x^4+x^2}\,dx$

$$\begin{array}{r|l} & x^2-1 \\ \hline x^4+x^2 & x^6 \qquad +2 \\ & -(x^6+x^4) \\ \hline & -x^4+2 \\ & -(-x^2-x^2) \\ \hline & x^2+2 \end{array}$$

$$\int x^2-1 + \frac{x^2+2}{x^2(x^2+1)}\,dx$$

$$\frac{A}{x}+\frac{B}{x^2}+\frac{Cx+D}{x^2+1} = \frac{x^2+2}{x^2(x^2+1)}$$

$$Ax(x^2+1)+B(x^2+1)+x^2(Cx+D)=x^2+2$$

$$Ax^3+Ax+Bx^2+B+Cx^3+Dx^2=x^2+2$$

$$x^3(A+C)+x^2(B+D)+Ax+B=x^2+2$$

$C=0 \quad B+D=1 \quad A=0 \quad B=2$

$D=-1$

$$\int\left(x^2-1+2x^{-2}-\frac{1}{x^2+1}\right)dx = \frac{1}{3}x^3-x-\frac{2}{x}-\tan^{-1}x+C$$

6. (5 points) Integrate $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx$

$m=\sqrt{x}$

$dm=\frac{1}{2\sqrt{x}}\,dx$

$2m\,dm = dx$

$$\int \frac{\cos m}{\cancel{m}}\left(2\cancel{m}\,dm\right)$$

$$2\int \cos m\,dm = 2\sin m + C = 2\sin\sqrt{x}+C$$